



Deep similarity segmentation model for sensor-based activity recognition

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Abstract

Signal segmentation is a critical stage in activity recognition. Most existing studies adopted the fixed-size sliding window method for this stage. However, the fixed-size sliding window may not produce the most effective segmentation method since human activities have variable length durations, particularly transitional activities. In this paper, we propose a novel deep similarity segmentation model that overcomes not only the limitations of the fixed sliding window method but also the weaknesses of threshold-based segmentation methods. Specifically, a novel deep learning model is designed to distinguish between transitional and basic activity by treating the segmentation task as a binary classification task. The proposed model accepts multiple sequence windows and extracts the local features automatically for each window using convolutional neural networks. The temporal features of windows are extracted by measuring the similarity and differentiation between the local features of adjacent windows. The local features are combined with the temporal features and passed to deep fully connected layers to distinguish the transitional activity from the basic activity windows. The evaluation relies on two public datasets, SBHARPT and FORTH-TRACE. According to the experimental findings, the proposed approach can distinguish between basic and transitional activities with an accuracy of 98.51% and 98.41%, respectively. Additionally, our method outperformed the fixed sliding window for activity recognition by 2.93% and 2.24% for both datasets, respectively, achieving an accuracy of 93.35% and 84.96%. These results are significant and outperform the precision of cutting-edge models.

Keywords Signal segmentation · Deep learning · Transitional activity

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1 Introduction

Human Activity Recognition (HAR) has recently emerged as one of the most significant research fields and a point of emphasis for researchers. HAR helps to understand human behaviors by recognizing human daily activities as well as it is used in a variety of Internet of Things (IoT) applications [1], healthcare [2], security, sports [3], and smart homes [4]. Human activities can be divided into two categories: basic and transitional (or postural transition) activities. The basic activities are those performed by a human in daily life, whether static or dynamic. The static activity does not contain motions during its performance (e.g., standing and sitting), while the dynamic activity contains motions during its performance (e.g., walking, jumping, and running). The transitional activities occur between two basic activities, such as stand-to-sit and sit-to-stand. There are two differences between the two categories. The first difference is that basic activities are performed for a long duration of time, whereas transitional activities are performed for a short duration of time. Second, basic activities occur more frequently than transitional activities. These variations may make it more challenging to identify and recognize transitional activities in comparison to basic activities [5, 6]. As a result, few studies included transitional activities in their HAR models. However, transitional activities are crucial in many applications, including activity tracking, healthcare, identifying falls in elderly people [7], and distinguishing the many consecutive activities [8]. Additionally, the transitional activities may introduce signal fluctuations if they are not taken into account by the HAR model, which would then affect the model's effectiveness.

Recently, inertial sensors like accelerometers and gyroscopes have become the most widely used embedded sensors in a variety of wearable technologies [9]. These sensors record a person's body movements while they engage in an activity. The gyroscope measures rotation along three axes, whereas the accelerometer measures acceleration along three axes.

In general, sensor-based HAR goes through three basic stages: signal segmentation, feature extraction, and classification stages. The signal segmentation stage separates the signals into several subsequent parts called windows. This is done by using the sliding window approach whereby each window contains a group of samples. The most commonly used signal segmentation method is the fixed-size sliding window [10]. In this method, the signal is segmented into fixed equal-sized windows. However, selecting an optimal window size is a challenging task due to the varying duration of human activities. A large window size might include data from different activities, whereas a small window size might not be able to include all of the data from a single activity. Furthermore, the fixed sliding window treats each window in isolation from the other windows, whereby the relationship between adjacent windows is not accounted during the classification. As a result, the developed model may not be able to correctly classify the windows. Thus, using the fixed window size for signal segmentation stage is not the most effective way to perform activity recognition, especially, those considered the transitional activity [5, 11, 12].

Recently, several studies proposed various signal segmentation methods to overcome the limitations of the fixed sliding window. Most of these studies are based on some threshold values with hand-crafted features. Although the deep learning techniques outperform traditional machine learning approaches for activity recognition [10, 13], few studies used a deep learning model in signal segmentation stage [10]. However, the aforementioned studies suffer from the following limitations: affected by noises in sensor data, difficulty in determining the optimal threshold for various activities, experiment dependent variables,

and performance regression when applied to practical scenarios [14]. In addition, most of these studies do not take into account the relationship between adjacent windows.

In this paper, we propose a novel Deep Similarity Segmentation (DSS) model for signal segmentation stage in HAR. The proposed model treats the segmentation task as a binary classification task that is able to distinguish between transitional from basic activity windows. The DSS model is built based on convolutional neural networks (CNN). The CNN layers allow the model to extract the features automatically to avoid hand-crafted features. Furthermore, the proposed model allows dependency on the learning-based model and avoids the limitations of threshold-based methods. Specifically, the signal is segmented first by using the fixed-size sliding window. A group of successive windows is fed to the proposed model. Then the local features are extracted by CNN layers for each window. The temporal features of windows are extracted by measuring the similarity of the features between adjacent windows. Local features are combined with temporal features and passed to the deep neural network layer to distinguish the transitional activity from the basic activity windows. Combining the local features with the temporal features enhanced the quality of the features and created robust features for the classification stage. For the activity classification task, two deep-learning-based models are built. The first aims to classify the basic activity, and the other aims to classify the transitional activity. Dividing the recognition task into two models allows each model to focus on learning its own set of activity classes.

The main contributions of the paper are summarized as follows:

- A novel deep learning similarity segmentation model is proposed, which treats the segmentation task as a classification task by fusing the local and temporal features to distinguish between transitional and basic activity. This overcomes the limitations of the fixed sliding window and threshold-based methods.
- Improving the quality of features for activity detection by proposing temporal features based on comparing the similarity of the features between adjacent windows and combining them with the local features of the window. HAR performance is improved by improving the quality of the features.
- The proposed segmentation model improved the segmentation task, which enhanced the performance of the HAR model and outperformed state-of-the-art HAR models that considered transitional activities.

The rest of the paper is organized as follows: Related work is the next section. The third section is explaining our proposed approach. Next, discusses the experiments and results. The final section is the conclusion.

2 Related work

Human activity is categorized into two categories: basic activities (BA) and transitional activities (TA). Humans engage in basic activities in daily life, whether they are long (such as walking) or short (such as jumping). A transitional activity occurs between two basic activities. Compared to the basic activity, the transitional activity differs from it in two key features: low duration time and occurrence (e.g., sit-to-stand). As a result, most researchers neglected the transitional activities in HAR.

However, signal segmentation is the first stage in HAR. Most studies used the fixed-size sliding window method for segmentation [15–17]. The drawback of this method is that a small window size could split a particular activity over several windows and provide less information about the activity. On the other hand, a large window size may mix signals from different activities, which causes overlap between them and increases noise. Some studies [3, 18] investigated the impact of window sizes on the signal segmentation process. Banos et al. [19] analyze and evaluate the effect of fixed window size on HAR model performance. It is found that, a short window size (0.25–0.5 s) with non-overlap is preferable to a large window size for recognition performance.

A dynamic window size is one solution to overcome the fixed-size sliding window limitations. In the dynamic sliding window, the sensor data is segmented into different sizes of windows based on specific features. Alhammad et al. [2] proposed a segmentation method that depends on the peaks and valleys of the Y-axis of the acceleration signal. The technique relies on selecting the valley's largest value and the peak with the lowest value. Then, a threshold value is defined to calculate the shortest distance between peaks. The activity boundaries are then determined based on the number of peaks in each activity. Two types of boundaries are suggested: peak and valley boundaries.

The peak boundary is activated if the activity has two peaks, which represent the boundary of the activity. If the activity has only one peak the valley boundary is activated which the two valleys that directly preceded and followed each peak were identified to represent the activity boundary. Akbari [20] suggested a hierarchical signal segmentation approach. The approach segmented the data using a large fixed-window size and extracted the features to predict the activity. Then, the window is divided into smaller sub-windows to precisely assign labels to each sub-window if it is believed to contain several activity signals. Li et al. [5] propose a signal segmentation approach by dividing the data into several segments. K-means clustering is used to categorize the segments once the features have been extracted from the windows. The approach combines the adjacent windows that belong to the same category. Using the windows duration period, transitional activity windows are distinguished from basic activity windows. Zhou et al [21]. introduced a segmentation model based on multi-probability thresholds. The sliding window method is used to divide the training dataset into windows depending on activity. In order to train the traditional classifiers, the features are manually extracted from the windows. Next, a threshold segmentation algorithm is employed to determine the optimal threshold settings based on the connection between peak and trough points. These parameters are used to locate the suspected static windows in signals, after segmenting the testing data using a sliding window. Additionally, the slope-area filtering technique is used to remove anomalous signal points. To determine the probability of each activity, a novel multi-label weighted probability model is proposed. The model can distinguish between the main activity and the unknown activity by combining the proposed segmentation results with the multi-probability threshold.

Furthermore, some studies present the segmentation method by detecting transitional activities. Pilario et al. [22] used the standard deviation trend analysis (STD-TA) technique to differentiate between basic activity windows and transitional activity windows. The signals are divided equally using the sliding window approach, and the statistical features of the segments are extracted. Then, the activity is classified using the SVM classifier. Noor [12] devised a method for adjusting the adaptive window size. Initial segmentation is carried out with a fixed window size, after which the window is divided into transitional activity and non-transitional activity. If the window represents a transitional activity, the adaptive sliding window

is enabled, and the window size is increased to accommodate more activity data based on the likelihood that the window belongs to a certain activity.

Recently, some studies have adopted the deep learning approach for the segmentation stage. A deep learning CNN-based system for activity segmentation was suggested by Xiao et al. [14]. Using the sliding window technique, the sensor data is split into groups of bins, and each group is then given a label. By noticing changes in the labels that occur in the set of bin labels, the network can determine when activity begins and ends. To minimize costly annotation efforts, He et al. [23] adopted a weakly supervised learning setting. The signal is partitioned into multiple windows. The kernel embedding of distributions is used to extract the features for each window and to learn a classifier for activity recognition. Signal segmentation, feature extraction, and activity classification tasks are performed jointly to update the learnable parameters of different components. Noor et al. [24] proposed a hybrid deep learning model that takes advantage of both temporal features and the relationship between windows in their HAR model by using the deep temporal Conv-LSTM architecture.

Additionally, some studies addressed the similarity concept in the segmentation task. Duan et al. [25] proposed a multi-task deep learning approach that jointly optimizes the activity recognition and segmentation tasks. The approach consists of three modules: the selective kernel, multi-scale window generation, and a prediction module. The first module generates the features, and the second module generates a certain number of multiscale windows from the features. In order to measure the similarity between the generated window and the truth activity bounding box, a Jaccard index is used. The prediction module consists of two branches: one for activity recognition and the other for activity segmentation. The training and prediction process is done by giving the module the activity labels for each window with the activity bounding box. However, the transitional activities and the relationship between windows are ignored in this study. Baraka and Noor [26] proposed a similarity segmentation approach to distinguish between basic and transitional activities. The approach measures the similarity between adjacent windows by using two steps. The first step divides the window into three windows and calculates their similarity by calculating the distance between them. The second step is calculating the similarity between the features of adjacent windows. By using threshold values, it can distinguish between basic and transitional activity windows. Although this method is too similar to the suggested model, it is threshold-based and uses feature engineering to quantify similarity, which necessitates additional research and domain expert to calculate threshold values.

In summary, most recent studies relied on handcrafted features with threshold values in their proposed segmentation methods. Only a few studies relied on the deep learning model. Furthermore, they treated each window independently, regardless of any temporal relationship between the adjacent windows. Whereas, most of the research that considered the temporal features used them in the activity classification stage, not in the segmentation stage [11, 24, 27].

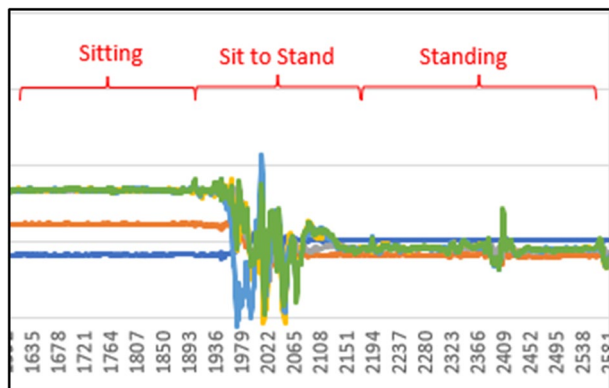
In this paper, we propose a deep learning similarity segmentation approach for the segmentation stage. The model extracts the features automatically by using CNN. Also, the temporal relationship between adjacent windows is taken into account to distinguish the transitional activity windows from the basic activity windows.

3 Proposed model

The fixed sliding window divides the signals from the wearable devices into a series of windows. Each window contains a group of data signals called samples. The number of samples collected in a single window is called the window size. i.e., $X = \{x^1, x^2, \dots, x^N\}$, where N is the total number of windows, $x^i \in \mathbb{R}^{w \times C}$, w is the window size, and C is the number of sensor's channels. The successive windows represent the time-varying nature of the activity signal over time. Therefore, there is a temporal relationship between the adjacent windows. If two adjacent windows belong to the same activity, their signal patterns are often the same. That means the similarity between them is high and the differentiation is slight. Contrary, if two adjacent windows belong to different activities, their signals' patterns are often different. That means the similarity between them is low and the differentiation is high. The deep similarity segmentation (DSS) model is proposed to exploit the similarity of the features between adjacent windows to distinguish between transitional activity and basic activity. The proposed model accepts a sequence of windows and measures the similarity between the features of these windows. If the window belongs to transitional activity, the similarity between it and adjacent windows is low and the differentiation is high. Figure 1 shows three different activity signals with different characteristics.

The overview of the proposed human activity recognition stages is given in Fig. 2. The proposed DSS model accepts a sequence of k windows segmented by the fixed sliding window. The DSS model classifies the window (x^{i-1}) into a basic or transitional activity window, in which x^i is the current window. Although classifying the current window is possible, we found that classifying x^{i-1} yields better performance. If the DSS model classifies the window as a transitional activity window, the window is concatenated with the previously classified transitional activity window. Then, the next sequence is fed to the DSS model for classification. If the window being classified is basic activity window, the window is passed to the basic activity classifier. The concatenated window is then classified by the transitional activity classifier. Since the size of the concatenated window is larger than the input size of the classifier, an interpolation method is applied to resize it to the size of the classifier's input.

Fig. 1 The difference between the signal characteristics of the basic and transitional activity



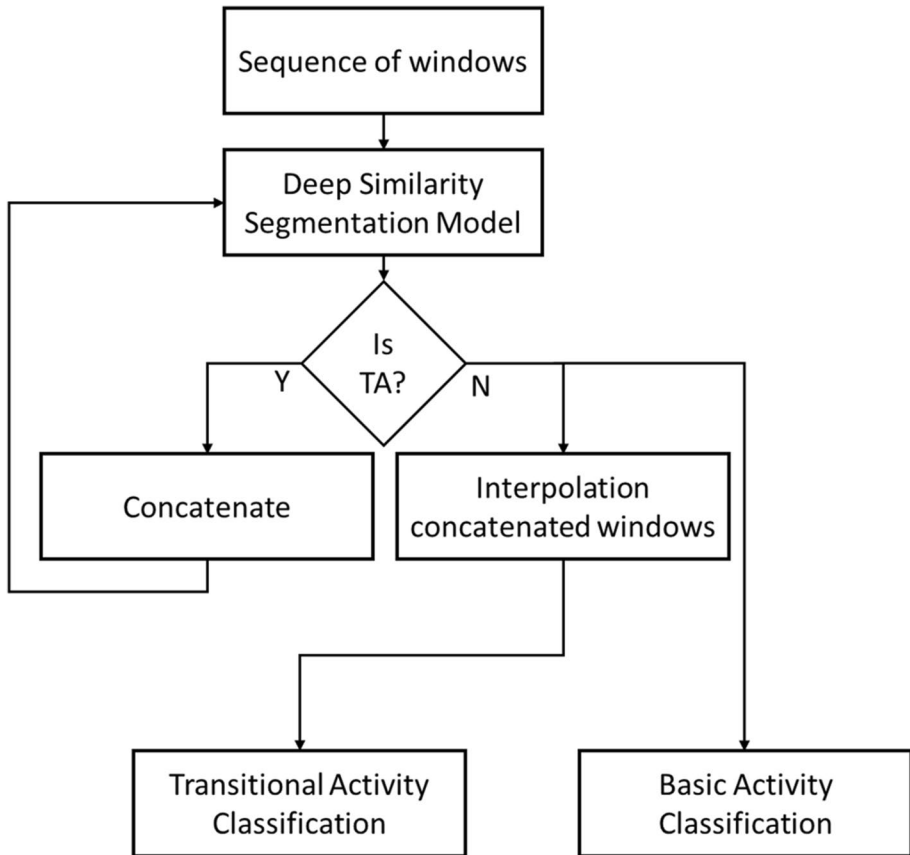


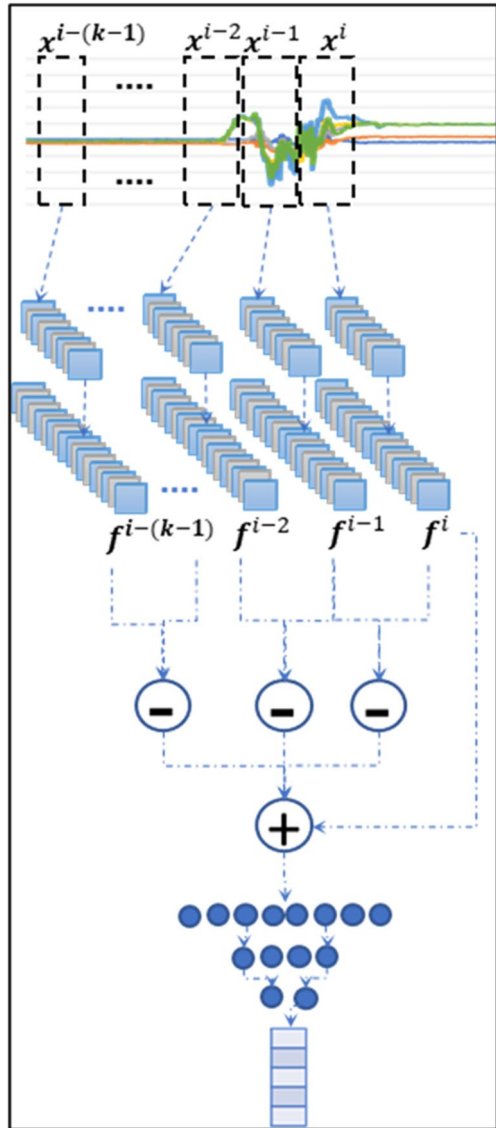
Fig. 2 Overview of proposed activity recognition approach

3.1 Deep similarity segmentation model

The DSS model is a deep learning model that aims to segment the signals by distinguishing between the basic and transitional activity windows. This is done by considering signal segmentation as a binary classification task. The DSS model architecture is illustrated in Fig. 3. The proposed model consists of k parallel feature learning pipelines to accept a sequence of windows i.e., $\mathbf{x}^{i-(k-1)}, \dots, \mathbf{x}^{i-2}, \mathbf{x}^{i-1}, \mathbf{x}^i$. Each feature learning pipeline consists of convolutional and max-pooling layers for extracting features from each window. All feature learning pipelines have the same architecture as shown in Table 1.

The DSS model extracts two types of features: local and temporal features. The local features of each window are extracted by the feature learning pipelines. The local features of each window from successive windows are indicated by $(f^{i-(k-1)}, \dots, f^{i-2}, f^{i-1}, f^i)$, where i refers to the number of windows (i.e., f^i refers to the local features of current window). The local features of the windows are then used to extract the temporal features. Given the local features of k windows, the temporal features are extracted by measuring the similarity between the adjacent windows. Finally,

Fig. 3 The DSS model architecture



the temporal features are combined with the local features of the window that are required to be classified as either a basic or transitional window.

We explain the formulation of the proposed DSS model with three input windows, $k = 3$, (i.e., x^{i-2}, x^{i-1}, x^i). First the model extracts the local features of the sequence of windows, i.e., f^{i-2}, f^{i-1}, f^i . The temporal features are extracted by measuring the similarity between local features for every two adjacent windows, i.e., between window (x^i) and (x^{i-1}), and between window (x^{i-1}) and (x^{i-2}).

Table 1 The architecture of the feature learning pipelines of DSS model

	Layer	Parameters	Output size
1	Input	Shape size = $w \times C$, where w is window size and C is sensor channels	-
2	Conv (ReLU)	Num. of filters = 512, kernel size = 3, stride = 1	$w - 2 \times 512$
3	Max Pooling	Pool size = 2, stride = 2, padding = 0, dilation = 1	$(w - 2)/2 \times 512$
4	Dropout	Rate = 0.5	$(w - 2)/2 \times 512$
5	Conv (ReLU)	Num. of filters = 1024, kernel size = 3, stride = 1	$(w - 4)/2 \times 1024$
6	Average Pooling	Pool size = 2, stride = 2, padding = 0, dilation = 1	$(w - 4)/4 \times 1024$
7	Dropout	Rate = 0.5	$(w - 4)/4 \times 1024$

Let f^i and f^{i-1} are the local features of windows (x^i, x^{i-1}) , respectively which extracted by the DSS feature learning pipelines. The similarity differentiations (S_1) are given as shown below.

$$S_1 = (f^i - f^{i-1}), \forall f^i \in \mathbb{R}^{v \times D}$$

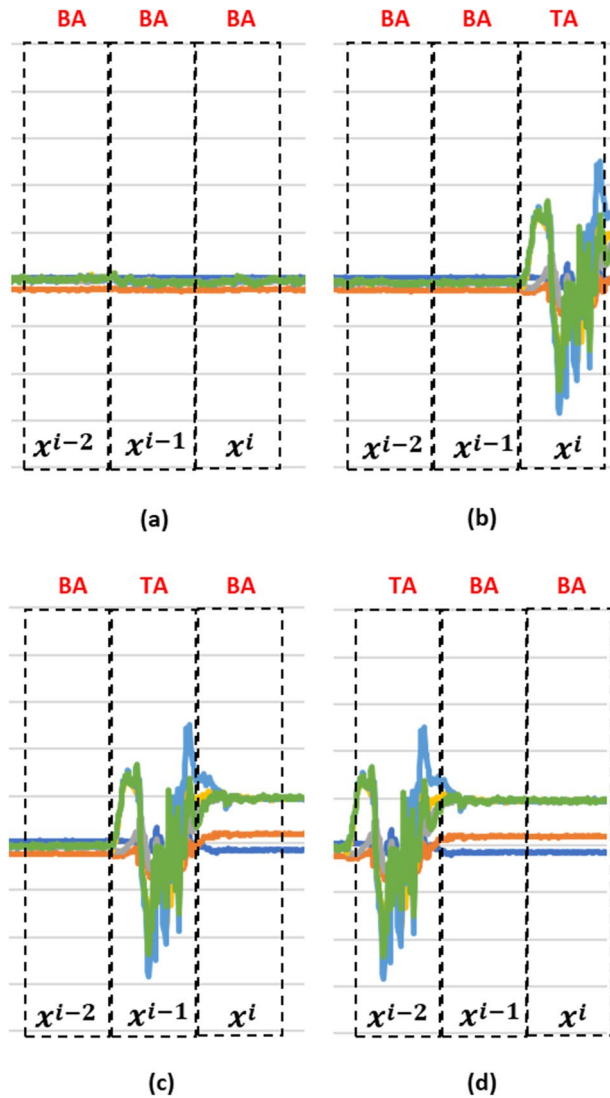
where v is the feature length (step size) and D is the depth of the features. Similar operation is applied to prior two adjacent windows (x^{i-1}) and (x^{i-2}) to measure the second similarity differentiation (S_2).

$$S_2 = (f^{i-1} - f^{i-2}), \forall f^{i-1} \in \mathbb{R}^{v \times D}$$

The temporal features between the adjacent windows describe the similarity between the features of the adjacent windows s_1 and s_2 . As shown in Fig. 4, the pattern of the transitional activities is significantly different from the basic activity. The difference between the window's features will be prominent if the classified window belongs to the transitional activity. In this case, we have four cases for type positions of the three adjacent windows, as illustrated in Fig. 4.

In the first case, all windows belong to the same basic activity, as shown in Fig. 4a. In this case, the similarity between windows is high and the two differentiation values are low. That means the differentiation values have little effect when combined with the window features. In case 2, the current window (x^i) belongs to transitional activity while the others belong to basic activity, as shown in Fig. 4b. The similarity between the two previous windows is high, whereas it is low between the current window and the one that follows it. So, while the first differentiation value s_1 is high, the second differentiation value s_2 is low. That means the differentiation values do not have a high effect when combined with the window features. In case 3, all windows belong to the basic activity except the middle window (x^{i-1}), which belongs to the transitional activity, as illustrated in Fig. 4c. The similarity between the three windows is low. So, both the differentiation values s_1 and s_2 are high. That indicates that when combined with the window features, the differentiation values have a significant impact. In the last case, as illustrated in Fig. 4d, the second previous window (x^{i-2}) belongs to transitional activity while the others belong to basic activity. The similarity between the two previous windows is low, whereas it is high between the current window and the one that follows it. Thus, the second differentiation (s_2) is high while the first differentiation value (s_1) is low. This indicates that when the differentiation values are combined with the window features, the differentiation values do not have a significant

Fig. 4 The four cases for the positions of the three adjacent windows. TA refers to the transitional activity, and BA refers to the basic activity window



impact. As a result, the similarity differentiation has a significant impact if the middle window (x^{i-1}) comes between two basic windows.

Following the temporal feature extraction, the model combines the window local features (f^{i-1}) with the two similarity features s_1 and s_2 by concatenating the three features together, as shown in the following equation.

$$F^{i-1} = g(S_1, f^{i-1}, S_2)$$

where g is the concatenation operation.

The concatenated features (F^{i-1}) represent the final features of the window (x^{i-1}). This combining of features enhanced the features quality of the window and makes it robust by considering the local features and temporal features. Then, the final features are passed

to the deep learning classification layers to classify the belonging of the window (x^{i-1}), whether the basic or transitional activity window.

3.2 Adjacent window concatenation and interpolation

Once the window is classified as basic or transitional activity, it is either passed to the activity classification model or kept for window concatenation. The window is passed to the basic activity model (in the activity classification stage) if it is classified as a basic activity window. Otherwise, the successive transitional activity windows are concatenated together and then combined into a single window to ensure all activity windows have the same size. The combination process is done using the linear interpolation method. The linear interpolation method combines multiple windows together while maintaining the signal patterns. The interpolation combination algorithm is illustrated in Algorithm 1.

Algorithm 1 The interpolation algorithm with concatenation process

x^n : An array contains the data of successive TA windows with shape $(n \times v \times D)$ where n is number of successive windows, v is the feature length (step size) and D is the depth of the features.
 c : Concatenated windows with shape $(n \times v \times D)$.
 r : An array with shape $(1 \times v \times D)$ is the result of the interpolation method on concatenated windows.
 L : Total features length of successive windows.
reshape(): Change the dimension of array shape.
concatenate(): combine multiple windows of the same shape.
linspace(): Return evenly spaced numbers over a specified interval.
interp1d(): Uses interpolation to find the value of new points.
stack(): Stack arrays in sequence horizontally (column wise).

```

1: for each window in  $x^n$ 
2:   if  $x^{i-1}$  is TA then
3:      $c = \text{concatenate}(c, x^{i-1})$ 
4:   else
5:      $L = n \times v$ 
6:      $x = \text{reshape } c \text{ from } (n \times v \times D) \text{ to } (L \times D)$ 
7:      $Ls1 = \text{linspace}(0, L, L)$ 
8:      $Ls2 = \text{linspace}(0, L, v)$ 
9:      $r = \text{new array.}$ 
10:    for each channel in  $D$ :
11:       $f = \text{interp1d}(Ls1, c \text{ data in channel, linear})$ 
12:       $r = \text{stack}(r, f(Ls2))$ 
13:    reshape  $r$  to  $(1 \times v \times D)$ 
14:    send  $r$  to TA classifier
15: send  $x^{i-1}$  to BA classifier

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Figure 5 shows an example of the result of applying the interpolation method to three adjacent transitional activity windows with shape $(n \times v \times D)$ where n is number of successive windows, v is the window size and D is the depth of the features. For example, consider six sensors' channels (3-axes for accelerometer and 3-axes for gyroscope), and three adjacent windows have been concatenated. Therefore, the input shape is $3 \times 60 \times 6$. Figure 5a shows the pattern of three adjacent transitional activity windows representing stand-to-sit activity, with each window has a size of 60 samples that present (i.e., the total size of the windows is 180 samples). Figure 5b shows the result of combining the three adjacent windows into a single window with a size of 60 samples using the interpolation method. The result shape is $(1 \times 60 \times 6)$. As shown, the interpolation

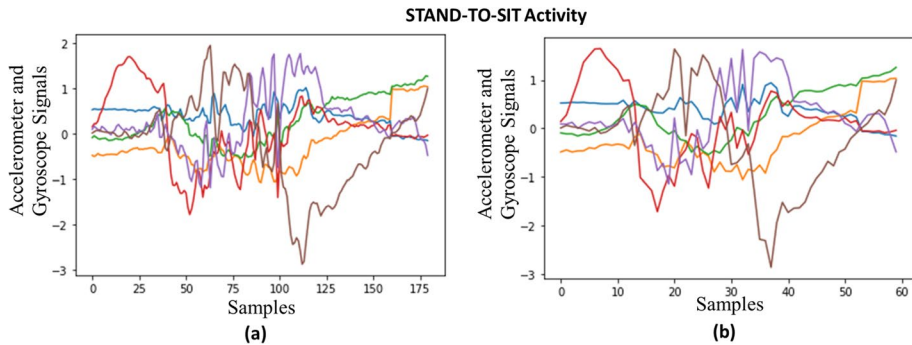


Fig. 5 The result of linear interpolation method to combine three adjacent transitional activity (Stand-To-Sit) windows (a) with shape $(3 \times 60 \times 6)$ into one window (b) with shape $(1 \times 60 \times 6)$

technique significantly preserved the pattern of the activity signal despite combining three windows into a single window. Experimentally, the features of the transitional activity become more distinctive and easier to discern when adjacent transitional activity windows are combined.

3.3 Activity classifiers

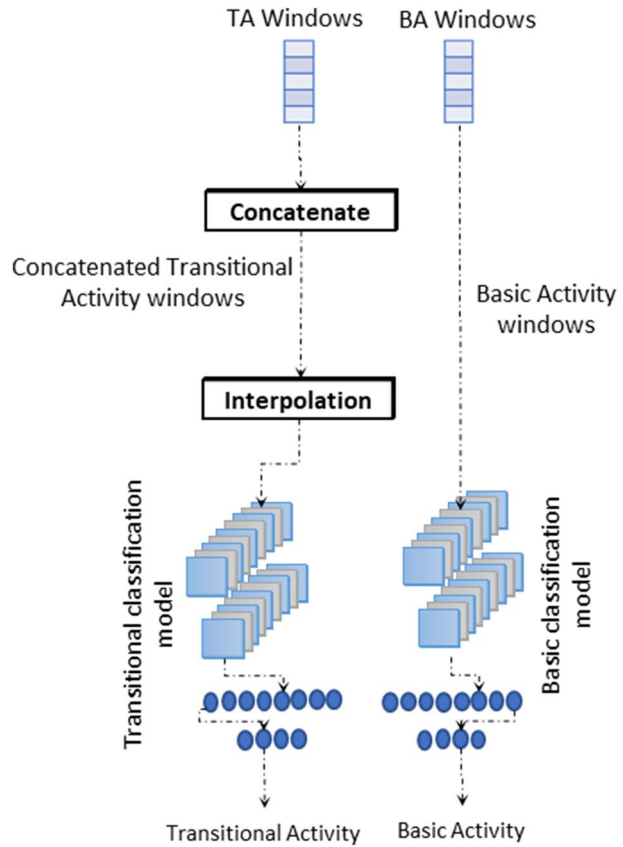
The classification model aims to classify the final result of human activity and consists of two models. The first model aims to classify the basic activities, while the second model aims to classify the transitional activities. The basic classification model accepts the basic activity windows directly from the DSS model. On the other hand, the transitional classification model accepts the transitional window passed from the interpolation method. Both models are based on convolutional, max-pooling and fully-connected layers, and are having the same architecture, as illustrated in Table 2.

The convolutional and max-pooling layers allow the model to extract the features automatically from each window for the classification task. Basic activity windows are classified using the basic activity model, while transitional activity windows are classified using the transitional activity model. Figure 6 shows the block diagram of the classification models.

Table 2 The architecture of the activity classifiers

	Layer	Parameter	Output size
1	Input	Shape size = $w \times C$, where w is window size and C is sensor channels	
2	Conv (ReLU)	Num. of filters = 512, kernel size = 3, stride = 1	$w - 2 \times 512$
3	Max pooling	Pool size = 2, stride = 2, padding = 0, dilation = 1	$(w - 2)/2 \times 512$
4	Dropout	Rate = 0.5	$(w - 2)/2 \times 512$
5	Conv (ReLU)	Input = 512, output = 1024, kernel size = 3, stride = 1	$(w - 4)/2 \times 1024$
6	Average pooling	Pool size = 2, stride = 2, padding = 0, dilation = 1	$(w - 4)/4 \times 1024$
7	Dropout	Rate = 0.5	$(w - 4)/4 \times 1024$
8	Flatten	start_dim = 1, end_dim = -1	$(w - 4)/4 \times 1024$
9	Fully connected	No. of units = 150, bias = True	150
10	Fully connected	No. of units = 50, bias = True	50
11	Fully connected	No. of units = 6, bias = True	6

Fig. 6 The block diagram of the activity classification model with the interpolation method, where TA refers to transitional activity windows and BA refers to basic activity windows



4 Experiments and results

This section explains the performance evaluation of the deep similarity segmentation approach. Additionally, we measure the effect of the deep similarity segmentation approach on human activity recognition performance.

4.1 Datasets description

One of the challenges facing in HAR research is the lack of standardized evaluation for HAR models and the lack of a standard benchmark dataset, which leads to an unfair comparison between the state-of-the-art HAR models [28, 29]. This is because building a benchmark dataset requires a large number of attributes that differ from one database to another, such as sensor types, sensor positions, device types, and activity types. This study aims to address the problem of signal segmentation involving continuous activity signals collected from a subject that performs the activities in continuous manners. Specifically, the dataset includes activity data of walking, standing, and sitting as well as the transitions between the basic activities such as stand-to-sit, sit-to-stand and sit-to-lie. However, according to our best knowledge, there are a few existing public datasets that contain activity data and labels of transitional activities [30]. The public datasets that meet the criteria

are SBHARPT [6] and FORTH-TRACE [31] datasets. SBHARPT is an updated version of the UCI HAR dataset [32] which consists of a collection of continuously recorded tri-axial signals obtained from accelerometer and gyroscope sensors. The smartphone's integrated accelerometer and gyroscope sensors are attached to the waist during performing the activities.

The dataset contains activity data from 30 participants who engaged in the following 12 activities: six basic activities, which included three dynamic activities (i.e., walking, walking downstairs, and walking upstairs) and three static activities (i.e., standing, sitting, and lying down), as well as six postural transitions between static activities, which included stand-to-sit, sit-to-stand, sit-to-lie, lie-to-sit, stand-to-lie, and lie-to-stand. The signals were recorded at a 50 Hz sampling rate. The dataset is divided by subjects into 23 subjects for the training stage and 7 subjects for the testing stage. We encoded the basic activities of walking, walking downstairs, walking upstairs, standing, sitting, and lying down as A1, A2, A3, A4, A5, and A6, respectively. In addition, we encoded the transitional activities of stand-to-sit, sit-to-stand, sit-to-lie, lie-to-sit, stand-to-lie, and lie-to-stand as T1, T2, T3, T4, T5, and T6, respectively.

The FORTH-TRACE dataset collects data from 15 participants wearing Shimmer sensor nodes in different places. A triaxial inertial sensor (accelerometer, gyroscope, and magnetometer) built within the sensor node recorded activity data at a fixed sampling rate of 51.2 Hz. Seven basic and nine postural transitional activities are performed by the participants. 11 participants are used for the training stage of the dataset's raw data, while 4 participants are used for the testing stage.

Furthermore, the FORTH-TRACE dataset contains talking-related activities such as sitting-while-talking. Since the accelerometer and gyroscope are primarily designed to capture human movements [33] and are not able to capture speech, both activities are treated in this study as belonging to the same class of activity [29]. Furthermore, some postural transitional activities appeared between two dynamic activities, which may have led to the misclassification of this posture activity. To prevent this, we considered just the first action, disregarding the second activity such as the climbing stairs to walk activity which became the climbing stairs activity.

We encoded the basic activities of standing, sitting, walking, and down/upstairs as A1, A2, A3, and A4 respectively. In addition, we encoded the transitional activities of stand-to-sit, sit-to-stand as T1 and T2 respectively.

4.2 Experimental setup

Initially, we perform some preprocessing to prepare the data signals to be fed into the proposed approach, such as cleaning the data and normalizing. The data cleaning aims to remove noise and irrelevant data from signals. The normalization process aims to scale the data signal.

The next step is segmenting the signal using the fixed-size sliding window method with non-overlapping into sequence of windows. The windows are the input for the DSS model. Since the minimum transitional activity duration time is approximately less than 2.6 s, the signals are segmented by three different window sizes of less than (2.6 s). Three distinct window sizes are employed in this study to assess the effect of window size on the effectiveness of the suggested approach. 60 samples (1.2 s), 90 samples (1.8 s), and 120 samples (2.4 s), respectively.

The dataset is divided by subject into training and testing datasets in order to train and test the proposed models.

Adam optimizer with a default learning rate (0.001) is used. Hyper-parameters are set empirically for the classification models. Also, a stochastic gradient descent optimizer is used for training the DSS model. The batch size is 64 windows. The maximum training epoch is 140 for training models. The implementation is performed by using Python, and the models build by using PyTorch on a workstation with Intel Core i5 CPU, 16G RAM, and GPU GTX1660.

4.3 Results and discussion

In this section, we explain and discuss the results and findings of the evaluation of the proposed approach. Three main types of experiments are conducted. The first experiment aims to measure the impact of the number of adjacent windows on the DSS model's performance. The segmentation DSS model's performance is assessed in the second experiment for different window sizes. The last experiment aims to evaluate the effectiveness of the HAR model using the DSS model for different window sizes. Additionally, we discuss how the DSS model affects the HAR model's efficiency in comparison to the fixed sliding window segmentation method.

The evaluation metrics used are overall accuracy, recall, precision, and f-score. The results of different activities are compared based on the f-score, which captures the results of the recall and accuracy measures.

4.3.1 Impact of the number of adjacent windows

The DSS model treats the segmentation process as a binary classification in terms of distinguishing between the transitional and non-transitional activity windows. To this end, the dataset is annotated as binary classes. The basic activity windows are annotated with 0, and the transitional activity windows are annotated with 1.

In the sliding window method, each window has two adjacent windows. So, the similarity features of each window are extracted by measuring the similarity between each window and its adjacent windows. In this section, we extract the similarity of different adjacent windows to measure the impact of the number of adjacent windows on the DSS model's performance. The experiment starts with two adjacent windows and is then increased to three and four, with the window size fixed at 90 samples. The results of the experiments are illustrated in Table 3. The accuracy of three and four adjacent windows is similar, while the accuracy of two adjacent windows is significantly lower, as shown in the table. Similar results are also observed for the f-score, which measures the different numbers of adjacent windows. Therefore, three adjacent windows are considered the optimal number of adjacent windows to extract a prominent similarity feature between the windows. Thus, our proposed model adopted three adjacent windows for activity recognition.

It is worth noting that the DSS model can classify any window that participated in extracting the temporal features by combining its local features with the temporal features. But the classification accuracy varies from one window to another. Experimentally, the middle window often obtains better results compared to other windows.

Table 3 The accuracy and f-score of three different adjacent windows for SBHART dataset

	Number of adjacent windows		
	Two	Three	Four
Accuracy	98.43%	98.51%	98.51%
F-score of basic activity	99.0%	99.0%	99.0%
F-score of transitional activity	89.0%	90.0%	90.0%

The best result is in bold

4.3.2 DSS model performance

This experiment aims to evaluate the performance of the proposed DSS model. We performed three experiments using three different window sizes (60, 90, and 120 samples) to measure the impact of the window size on the DSS performance. The performance of the segmentation approach is evaluated in terms of accuracy, precision, recall, and f-score metrics.

The accuracy testing results for three different window sizes are shown in Table 4. Based on the results, the proposed DSS model achieves a higher degree of accuracy in distinguishing between transitional and basic activities.

The window size of 90 samples achieves the best accuracy of 98.51% compared to other window sizes for the SBHART dataset. Also, the window size of 120 samples achieves the best accuracy of 98.41% compared to other window sizes for the FORTH-TRACE dataset. Moreover, the precision, recall, and f-score of each activity type are shown in Table 5.

The results show that the proposed DSS model achieved better performance in classifying the basic activities since the number of windows of transitional activities is less

Table 4 The accuracy of proposed DSS model to distinguish between transitional and non-transitional activities for both datasets with three different window sizes

Dataset	Window size		
	60 samples	90 samples	120 samples
SBHARPT	97.84%	98.51%	98.01%
FORTH-TRACE	97.78%	98.1%	98.41%

The best result is in bold

Table 5 The precision, recall and f-score of proposed DSS model to distinguish between transitional and non-transitional activities for both datasets with three different window sizes

Activity	SBHARPT			FORTH-TRACE		
	Precision	Recall	F-Score	Precision	Recall	F-Score
Window size = 60 samples						
Basic	99.0%	99.0%	99.0%	98.0%	99.0%	99.0%
Transition	87.0%	83.0%	85.0%	69.0%	46.0%	55.0%
Window size = 90 samples						
Basic	99.0%	99.0%	99.0%	99.0%	100.0%	99.0%
Transition	89.0%	91.0%	90.0%	77.0%	51.0%	61.0%
Window size = 120 samples						
Basic	100.0%	98.0%	99.0%	99.0%	99.0%	99.0%
Transition	80.0%	97.0%	87.0%	76.0%	66.0%	71.0%

The best result is in bold

than the number of basic activity windows. For SBHART dataset, the recorded f-scores are 99.0% and 90.0% for the basic and transitional activities, respectively, for window size equal to 90 samples. For FORTH-TRACE dataset, the f-scores are 99.0% and 71.0% for the basic and transitional activities, respectively, for window size equal to 120 samples. In summary, the results show that the proposed DSS model is able to effectively distinguish between the basic and transitional activities for both datasets.

4.3.3 Human activity recognition

This experiment measures the impact of DSS model on the HAR performance compared with a fixed-size sliding window. The experiments are performed using three different window sizes: 60, 90, and 120 samples. The results of activity recognition accuracy for the DSS model and fixed sliding window for both datasets are shown in Table 6.

For the SBHARPT dataset, it is observed that the DSS model achieved higher recognition accuracy for the 60 and 90 sample window sizes compared with the fixed sliding window. The fixed-size sliding window obtains more accuracy than the DSS model in window sizes of 120 samples due to the low performance of the DSS model and transitional activity model in this window size. The fixed sliding window achieved an accuracy of 89.75%, 90.42%, and 91.2% for window sizes of 60, 90, and 120 samples, respectively, while the proposed DSS model achieved an accuracy of 90.22%, 93.35%, and 86.75% for window sizes of 60, 90, and 120 samples, respectively. Thus, the DSS model improves the accuracy by 0.47% and 2.93% for 60 and 90, respectively. In addition, the results show that a window size of 90 samples achieved the best recognition accuracy, which is comparable to an existing study reported in [3]. Therefore, the following comparative analysis and discussion are based on the 90 samples window size.

Table 7 shows the precision, recall, and f-score measures of each activity for both the DSS and fixed sliding window for the SBHARPT dataset. Both the DSS model and the fixed sliding window achieved high performance in classifying the basic activities, with an f-score measure above 95% for all activity classes except A4 and A5 (i.e., sitting and standing). This is because both the sitting and standing activities are static and have close features. Overall, the average f-scores of the basic activities for fixed sliding windows and DSS are 93% and 95%, respectively.

As for transitional activities, the fixed sliding window performed poorly in classifying the activities, with the f-score measures in the range of 61–74%. That means the fixed-size sliding window poorly handles the transitional activities due to their varying lengths. Conversely, the DSS model achieved a significantly better performance in classifying the

Table 6 Comparison of HAR performance in terms of accuracy between two segmentation methods: fixed sliding window and DSS model with different window sizes for both datasets

	Window size					
	60		90		120	
Dataset	Sliding window	DSS	Sliding window	DSS	Sliding window	DSS
SBHARPT	89.75%	90.22%	90.42%	93.35%	91.2%	86.75%
FORTH-TRACE	80.83%	82.88%	82.06%	84.5%	82.72%	84.96%

The best results is in bold

Table 7 Activity recognition performance of fixed sliding window and proposed DSS model for SBHARPT dataset for a window size of 90 samples

Activity	Fixed Sliding Window			DSS		
	Precision	Recall	F-Score	Precision	Recall	F-Score
A1	96.0%	99.0%	97.0%	98.0%	99.0%	98.0%
A2	99.0%	92.0%	96.0%	98.0%	98.0%	98.0%
A3	93.0%	97.0%	95.0%	97.0%	98.0%	98.0%
A4	81.0%	91.0%	86.0%	85.0%	90.0%	88.0%
A5	90.0%	80.0%	85.0%	91.0%	83.0%	87.0%
A6	98.0%	99.0%	98.0%	99.0%	99.0%	99.0%
BA Average	93.0%	93.0%	93.0%	95.0%	95.0%	95.0%
T1	69.0%	55.0%	61.0%	96.0%	76.0%	85.0%
T2	69.0%	78.0%	73.0%	83.0%	87.0%	85.0%
T3	68.0%	55.0%	61.0%	73.0%	95.0%	83.0%
T4	68.0%	81.0%	74.0%	72.0%	94.0%	82.0%
T5	49.0%	58.0%	53.0%	79.0%	75.0%	77.0%
T6	71.0%	63.0%	67.0%	93.0%	74.0%	83.0%
TA Average	66.0%	65.0%	65.0%	83.0%	84.0%	83.0%

transitional activities, with the f-score measures in the range of 77–85%. The proposed DSS model successfully detects the transitional activity windows and aggregates them to produce more effective window features. Overall, the average f-scores of the transitional activities for fixed sliding windows and the DSS model are 65% and 83%, respectively. Figure 7 shows the confusion matrix for human activity recognition at window size 90 using the proposed the DSS model.

For the FORTH-TRACE dataset, the DSS model achieved higher recognition accuracy for all window sizes of 60, 90, and 120 sample window size compared with the fixed sliding window. The fixed sliding window achieved an accuracy of 80.83%, 82.06%, and 82.72% for window sizes of (60, 90, and 120 samples), respectively, while the proposed DSS model achieved an accuracy of 82.88%, 84.5%, and 84.96% for window sizes of 60 samples, 90 samples, and 120 samples, respectively. Thus, the DSS improves the accuracy by 2.05%, 2.44%, and 2.24% for the 60, 90, and 120 samples, respectively. In addition, the window size of 120 samples achieved the best recognition accuracy.

In addition, Table 8 shows the precision, recall, and f-score measures of each activity for both the DSS model and fixed sliding window in the FORTH-TRACE dataset. As a result, both the fixed sliding window and the DSS achieved a moderate and close performance in classifying the basic activities, except for the standing activity. Similar to the SBHARPT dataset, the sitting and standing activities have a lower f-score compared with other activity classes. Nevertheless, the DSS achieved a better f-score measure for all activity classes compared with the fixed sliding window. Overall, the average f-scores of the basic activities for fixed sliding windows and the DSS model are 83.0% and 85.0%, respectively. The DSS model improved the performance of the basic activity by 2%.

As for the transitional activities, the fixed sliding window performed poorly in classifying the transitional activities, with an f-score measuring 33% and 24% for both transitional activities, respectively. That is, due to the varying lengths of the transitional

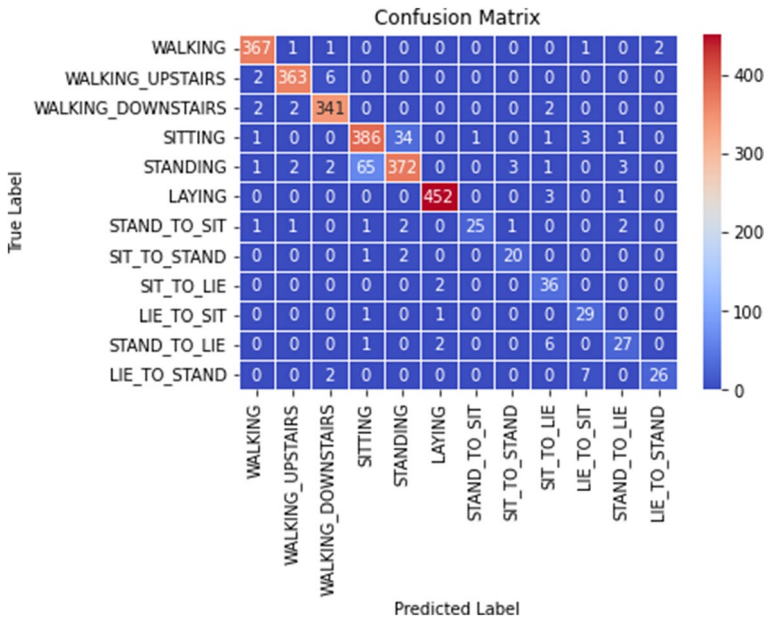


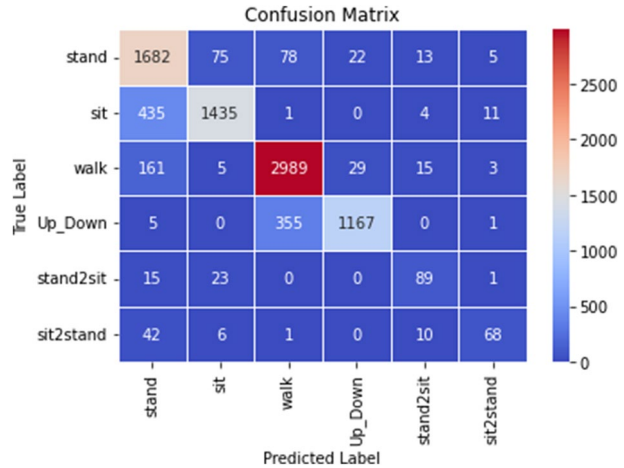
Fig. 7 Confusion matrix of activity recognition results using DSS model with window size of 90 for SBHARPT dataset

Table 8 Activity recognition performance of fixed sliding window and proposed DSS model for FORTH-TRACE dataset for a window size of 120 samples

Activity	Fixed Sliding Window			DSS		
	Precision	Recall	F-Score	Precision	Recall	F-Score
Stand	71.0%	80.0%	75.0%	72.0%	90.0%	80.0%
Sit	87.0%	79.0%	83.0%	93.0%	76.0%	84.0%
Walk	85.0%	93.0%	89.0%	87.0%	93.0%	90.0%
Climb Stairs	93.0%	80.0%	86.0%	96.0%	76.0%	85.0%
BA Average	84.0%	83.0%	83.0%	87.0%	84.0%	85.0%
Stand-To-Sit	65.0%	22.0%	33.0%	68.0%	70.0%	69.0%
Sit-To-Stand	37.0%	17.0%	24.0%	76.0%	54.0%	63.0%
TA Average	51.0%	19.5%	28.5%	72.0%	62.0%	66.0%

activities, the fixed window size cannot handle them. Furthermore, the proposed DSS model achieved better performance for classifying the transitional activities, with an f-score measuring 69% and 63% for both transitional activities, respectively. This low performance is due to combining the talking activity with its corresponding transition activity. Overall, the average f-scores of the transitional activities for fixed sliding windows and the DSS model are 28.5% and 63%, respectively. The DSS model improved the performance of the transitional activity by 34.5%. Figure 8 illustrates the confusion matrices of activity recognition using the DSS approach for 120 window size.

Fig. 8 Confusion matrix of activity recognition results using DSS model with window size of 90 for FORTH-TRACE dataset



4.3.4 Comparison with the state-of-art approaches

One of the major concerns with HAR is the lack of a benchmark dataset and a consistent way to evaluate the performance of the HAR model, which makes it unfair to compare state-of-the-art approaches [21]. However, we compared the performance of our proposed approach with that of some state-of-the-art models that use the SBHARPT database and consider transitional activity. Because it is the most commonly used dataset in studies that consider transitional activity.

The accuracy of the most recent HAR models on the SBHARPT dataset and segmentation technique is shown in Table 9 for each research. Due to the special's settings or constraints of each research, the comparison is not completely fair. Chen et al. [34] categorized all different kinds of transitional activities into one class, using the fixed sliding window as a segmentation tool. In conclusion, all prior models employed a fixed sliding window for segmentation, which had an impact on the model's performance due to the method's drawbacks. As indicated, the fixed-sliding window is ineffective for distinguishing between different activity types since it is challenging to choose the optimal window size for each type of activity. Zhou et al. [21] segmented the signals based on the probability threshold. The optimal threshold is obtained by hand-crafting features. The probability of each activity is obtained using a multi-label weighted probability model. Then, combining the segmentation results with a multi-probability threshold recognition in order to distinguish the useless segments in the complex continuous signals. Therefore, the segmentation stage is based on a hand-crafted setting. Li et al. [5] used the fixed sliding window approach to segment the data and K-Means cluster analysis to categorize the windows. The approach cannot be used in real-time systems because it requires continuous sensor data to be examined and clustered before the signal can be segmented, but in real-time systems the incoming sensor data must be segmented straight away. The model also learns a signal pattern that differs from the actual signal pattern of the activity signal as a result of repeated data.

In addition, Irfan et al. [35], Noor et al. [24], and Yulita [36] use the fixed-size sliding window method for the segmentation stage. The statistical features matrix is used for each activity instead of the raw sensor data. Noor et al. [37] utilized an adaptive sliding window for the segmentation stage. The adaptive window depends on hand-crafted

Table 9 Summary of the state-of-the-art segmentation methods and the proposed segmentation model

Paper	Accuracy	Segmentation method	Description
Chen et al. [34]	98.00%	Fixed sliding window	Limitations of Fixed sliding window. A large window size might include data from different activities, whereas a small window size might not be able to include all data of a single activity.
Zhou et al. [21]	97.71%	Multi-probability Threshold	The relationship between windows is neglected and requires manual setting and experiments to determine the optimal value of the threshold.
Li et al. [5]	97.34%	Clustering-based	The proposed segmentation method is not applicable in real-time setting.
Irfan et al. [35]	96.11%	Fixed sliding window	Limitations of Fixed sliding window.
Noor et al. [37]	93.40%	Adaptive sliding window	The relationship between windows is neglected and requires an assumption about the data distribution.
Lone et al. [38]	93.05%	Fixed sliding window	Limitations of Fixed sliding window.
Noor et al. [24]	91.60%	Fixed sliding window	Limitations of Fixed sliding window.
Yulita [36]	86.00%	Fixed sliding window	Limitations of Fixed sliding window.
SSA [26]	92.71%	Similarity-based	Limitations of threshold based. Required additional researchs and domain expert to calculate threshold values, distinct performance declines for mixed activities, and it neglects the relationship between windows segmentation and recognition processes.
Proposed model	93.35%	Deep-Similarity-based	

features. Also, the window is enlarged without taking into consideration the temporal relationship between neighboring windows. In [38] the basic and transitional activity data are isolated and then recognized each activity independently. The segmentation technique used is the fixed sliding window. Additionally, many machine learning techniques are developed to get the best accuracy for each set. Furthermore, SSA [26] it is still threshold-based approach. The limitations of the threshold-based methods are that distinct performance declines for mixed activities, and they neglect the relationship between windows segmentation and recognition processes. In addition, it required experience and experimentation to detect the optimal value of the threshold.

As a result of the foregoing, most models employ hand-crafted features for segmentation. Also, each research owns has a special setting that is different from the others. This makes the comparison with each other accurate. Consequently, when compared to the results of state-of-the-art HAR models, the results of the proposed approach are significant and valuable.

5 Conclusion

In this paper, a deep similarity segmentation model is proposed for the signal segmentation process. The proposed model improved the quality of features by combining temporal features with local features to distinguish between transitional and basic activities. First, the local features of windows are extracted, and then the temporal features are extracted by measuring the similarity between the local features of adjacent windows. Both local and temporal features are extracted based on CNNs. The proposed model classifies the windows as either basic or transitional and concatenates the successive transitional windows. The interpolation method is performed on successive transitional windows to combine them into a single window. For the classification task, two CNNs classification models are proposed to classify the basic and transitional windows. Two public datasets are used to evaluate our model. The results show that our model enhanced the HAR model's performance compared with the fixed sliding window. It also obtains significant performance compared with the state-of-the-art HAR models. For future work, combining the segmentation model with the classification model to build an end-to-end model could improve the HAR model's performance.

Data Availability The datasets uses in this research during and/or analyzed during the current study are available in the UCI Machine Learning Repository, <https://doi.org/10.24432/C54G7M>, and Zenodo Repository, <https://zenodo.org/doi/10.5281/zenodo.841300>.

Declaration

Data sharing Data sharing is not applicable to this article, as no new data were created or analyzed in this study. This work uses two public datasets (SBHARPT and FORTH-TRACE) available at UCI Machine Learning Repository and Zenodo Repository.

Conflict of interest Abdulrahman Baraka and Mohd Halim Mohd Noor declare that they have no conflict of interest.

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